Roll No. | | | | | | | |

ODD END SEMESTER EXAMINATION DECEMBER - 2024

Name of the Course: **B.Tech CSE** Semest-.. **3rd**

Name of the Paper: **Logic Design & Computer Organization** Paper Code. **TCS 308**

Time: 3 Hours Maximum Marks: 100

Note:-

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks.

Q1	(20marks)	
(a)	Define essential prime implicants. Minimize the following Boolean expression using tabular method. $Y(A,B,C,D)= \sum m\ (5,7,8,10,13,15)+ \sum d\ (0,1,2,3)$	CO1
(b)	Implement the following function using active low decoder and 8:1 Mux. $F\ (A,\ B,\ C) = \pi M\ (0,\ 1,\ 2,\ 3,\ 4)$	
(c)	Construct 8 to 3 encoder using 2 to 1 encoder only.	
Q2	(20 marks)	
(a)	Determine the characteristic equation of JK flip flop. Construct a JK flip-flop using a D flip-flop, a 2:1 multiplexer and an inverter.	CO2,
(b)	Draw the circuit of universal shift register. What are the applications of shift registers?	CO3
(c)	Draw and explain the circuit of MOD 18 ripple counter.	
Q3	(20 marks)	
(a)	Design MOD 16 synchronous counter using T flip flop.	CO2, CO3
(b)	Draw and explain the operation of MOD 8 twisted ring counter. Also determine total number of unused states.	
(c)	Design a clocked sequential circuit with two D flip flops, A and B; one input x and one output Z is specified by the following next state and output equations: A (t+1) =Ax + Bx, B (t+1) = (A+B')x, Z=A+B. (a) Draw the logic diagram of the circuit. (b) Determine type of machine (Mealy or Moore)? (c) Draw state table and state diagram.	
Q4	(20 marks)	
	A program runs in 20 seconds on computer A, which has a 3 GHz clock	

(a)	rate. We are trying to help a computer designer build a computer, B, which will run this program in 12 seconds. The designer has determined that a substantial increase in the clock rate is possible, but this increase will affect the rest of the CPU design, causing computer B to require 1.2 times as many clock cycles as computer A for this program. What cloc'. rate should we tell the designer to target?	CO4, CO5
(b)	Draw the flowchart of booth's algorithm. Perform following signed number multiplication operation: -5 x 6.	
(c)	What do you mean by pipelining? Suppose time taken to process a sub-operation in each segment is 20 ns. Assume that there is 4 stage pipelining and there are total 100 tasks, then what will be speed up of that system?	
Q5	(20 marks)	
(a)	A computer has L1 and L2 caches. The hit rate is 0.95 for both caches. Time needed to access L1 cache is 1 cycle, L2 cache is 10 cycles and main memory is 50 cycles. Calculate the average access time experienced by the processor.	CO4, CO6
(b)	Describe memory hierarchy based on parameters like speed, size, complexity, and cost.	
(c)	Describe the Flynn's classification using neat diagrams. Also explain I/O interface.	